UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

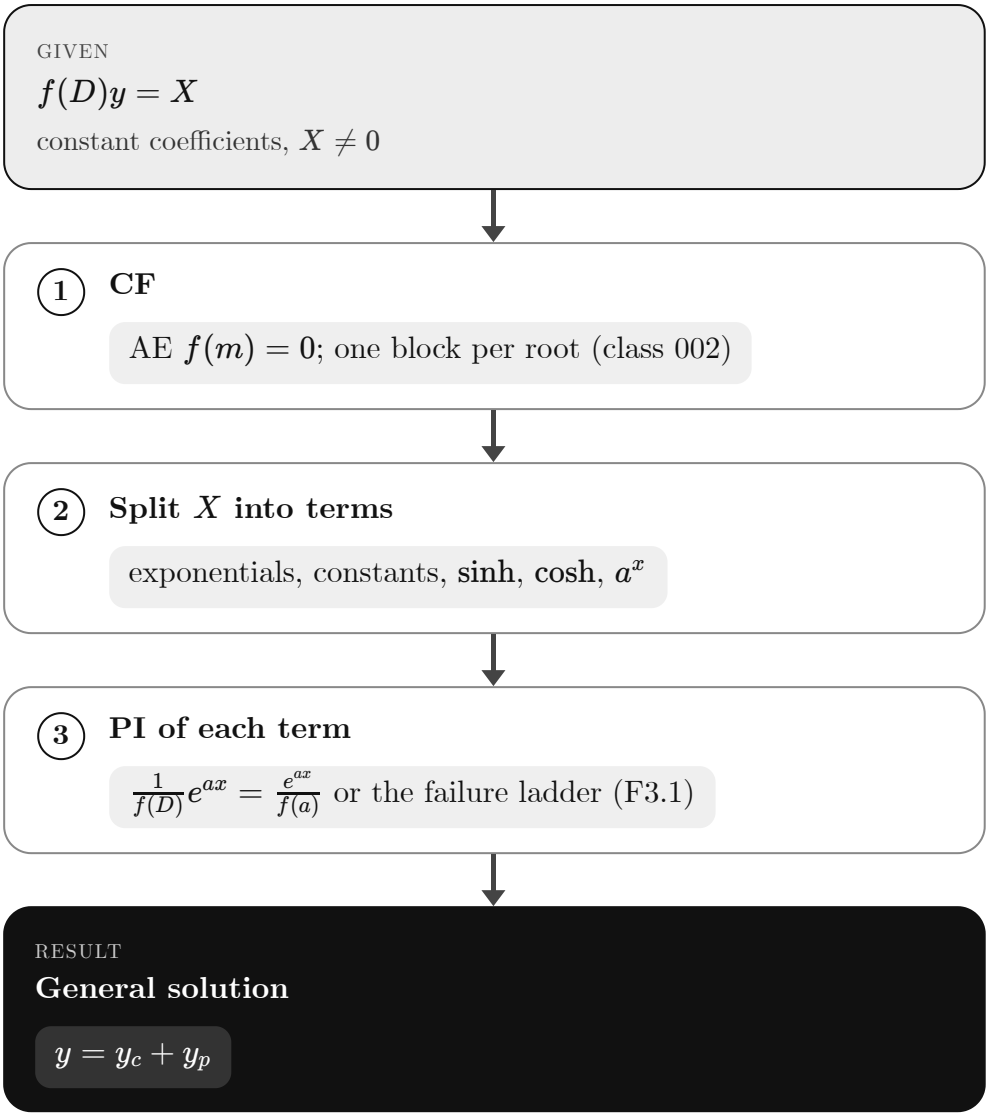
Class 003 · Particular integral Type 1 ($X = e^{ax}$) · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·8	Unit 2 deck 003 (“Particular integral of Type 1”), slide 8 (rendered .pptx)
T003 00:10:15	Unit 2 AV-summary transcript 003 (“Particular integral of standard functions – Type 1”, 21 min), segment start
T004a 00:02:06	Unit 2 AV-summary transcript in bucket 004 (“Particular integral Type 1 and Type 2”, 12.6 min); only its Type 1 part (00:00:00–00:06:04) and the constant term of its $\sin^2 x$ example are cited here. The class 004 pack cites the same file.
CW U2 P10	Unit 2 class-work problem 10
ESA Jan-24 s1 2d	Past paper: UE23MA141A Jan 2024 set 1, question 2.d

Contents

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3. Figure F3.1 · The Type 1 ladder for $\frac{1}{f(D)}e^{ax}$
4. Figure F4.1 · What replaces D?
5. Figure F4.2 · Two terms, two routes: $\frac{2\sinh x}{D^3-3D+2}$ (slides 003·16–17)
6. Type 1 and its cases of failure
7. Right-hand sides reducible to Type 1
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1 The inverse differential operator and the particular integral

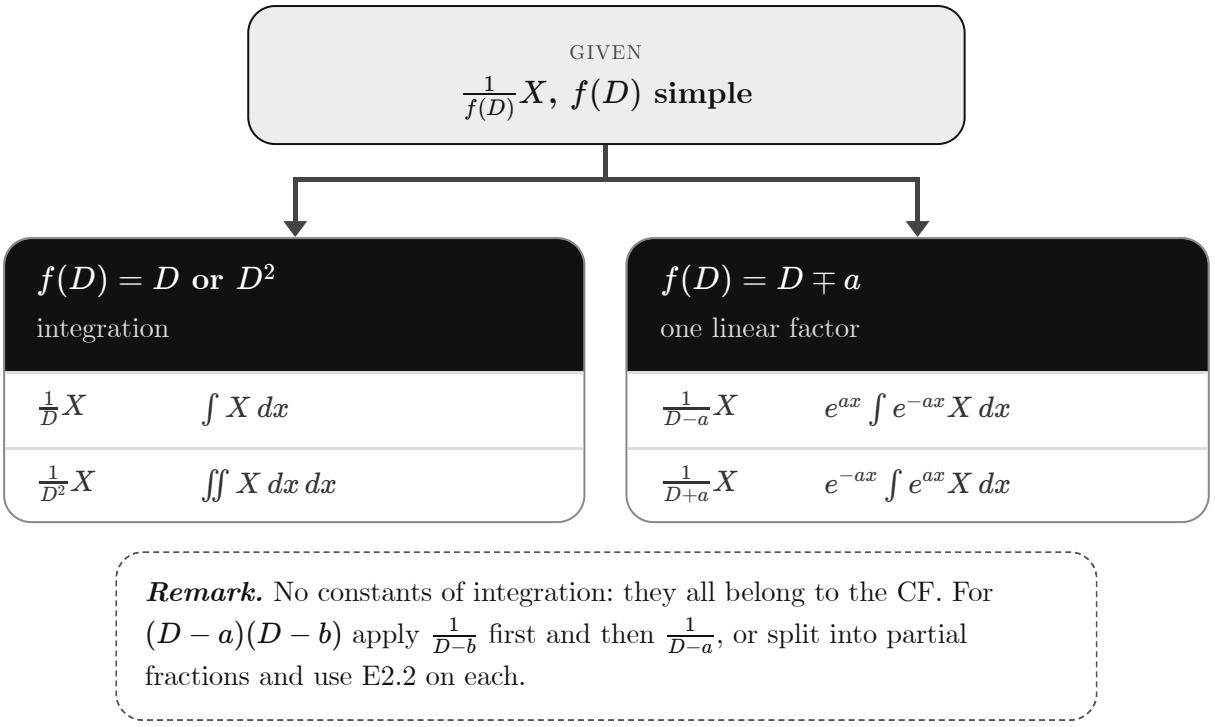


Remark. The CF roots tell you in advance which terms of X will fail: any e^{ax} whose a is an AE root.

Figure F1.1. Solving $f(D)y = X$ ESA ★★★

S003·6 · S003·11 · S003·12 · T003 00:17:57 · T003 00:20:36

2 Important results: $\frac{1}{D}$ and $\frac{1}{D-a}$

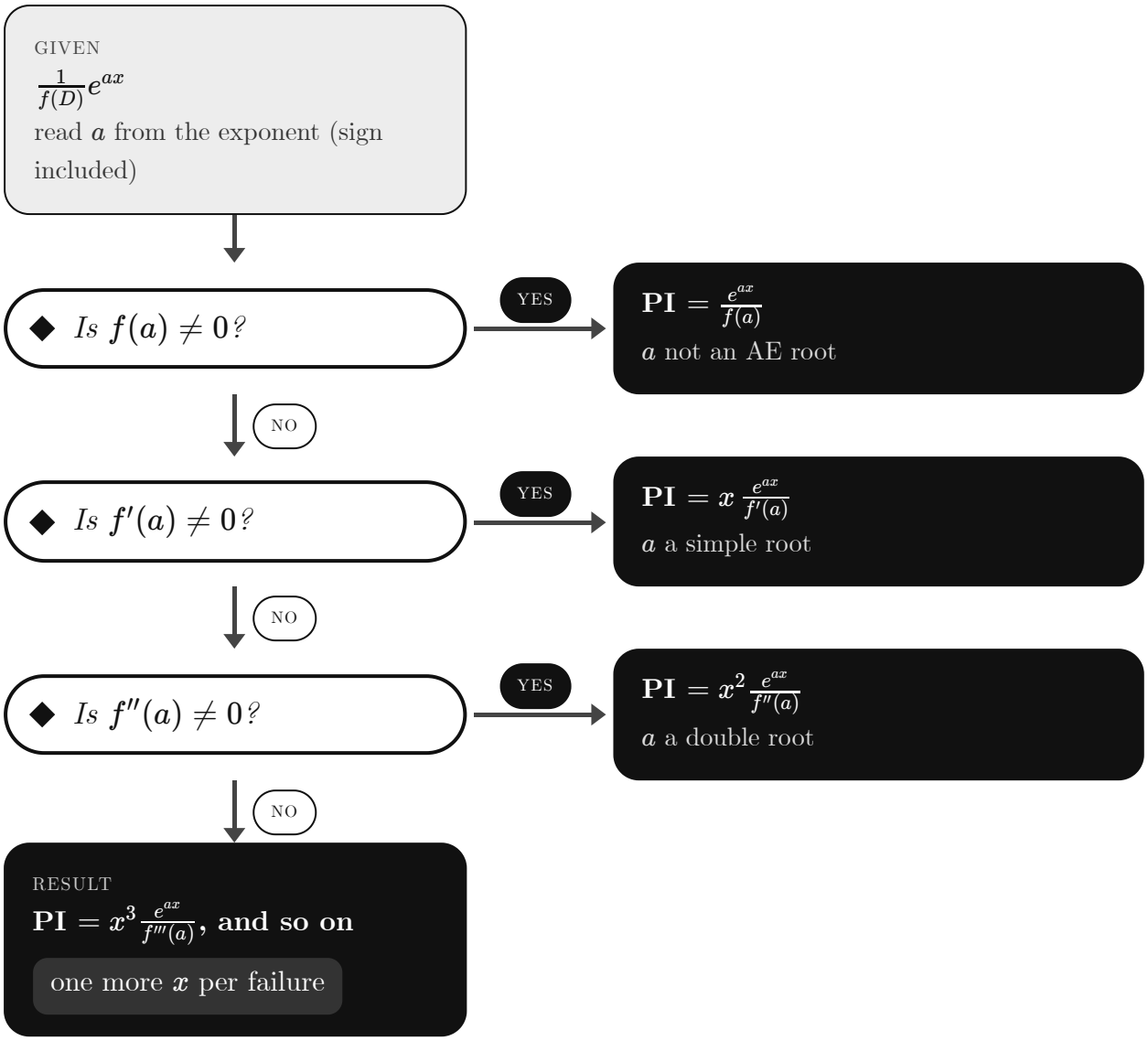


Remark. No constants of integration: they all belong to the CF. For $(D-a)(D-b)$ apply $\frac{1}{D-b}$ first and then $\frac{1}{D-a}$, or split into partial fractions and use E2.2 on each.

Figure F2.1. Important results for simple $f(D)$ ESA ★★

S003·7 · T003 00:06:24 · T003 00:07:29 · T003 00:09:02

3 Type 1: $X = e^{ax}$ and the cases of failure



Remark. Derivatives are taken with respect to D, then $D = a$. Shortcut (added for clarity): $f(D) = (D-a)^k \varphi(D)$ gives $\frac{x^k e^{ax}}{k!\,\varphi(a)}$.

Exam trap. Keep the constant factor: $\frac{8e^{2x}}{(D-2)^2} = 4x^2e^{2x}$.

Figure F3.1. The Type 1 ladder for $\frac{1}{f(D)}e^{ax}$ ESA ★★★

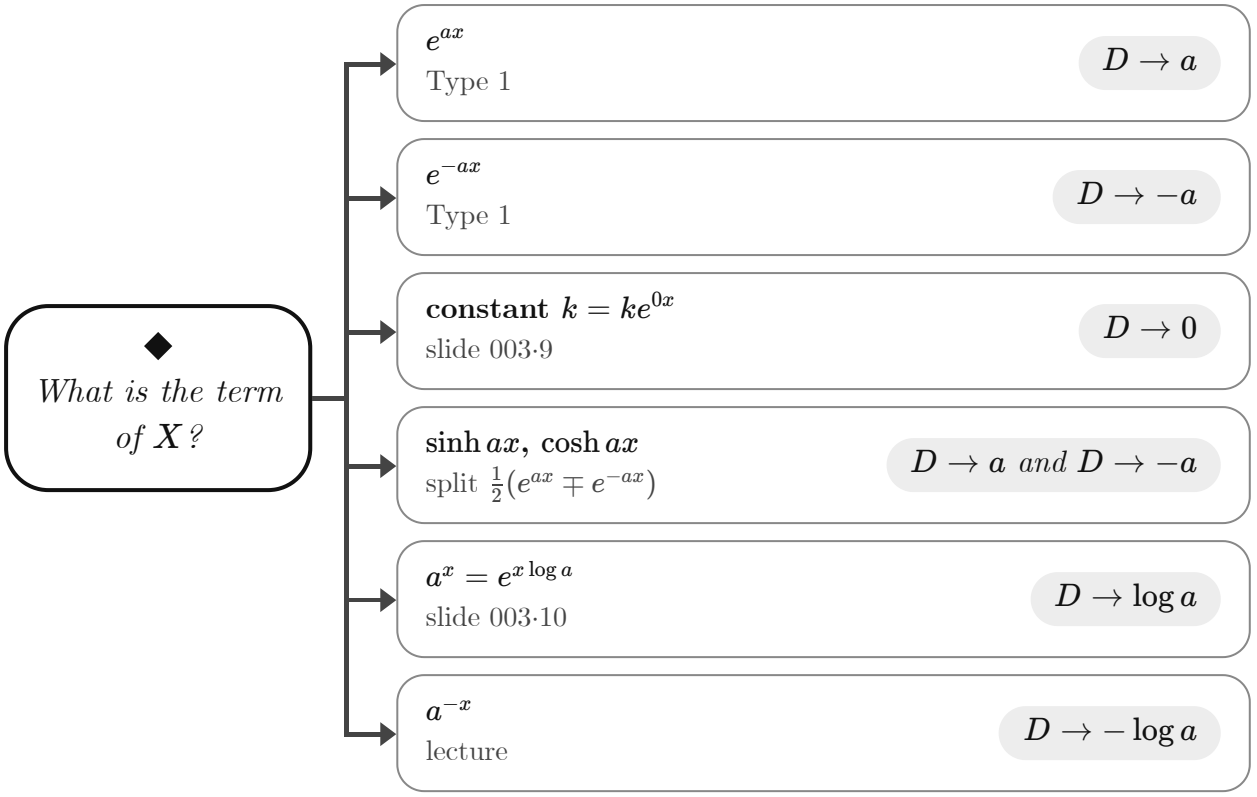
S003·8 · S003·13 · T003 00:11:35 · T003 00:12:53 · T003 00:13:19 · T004a 00:01:01

Type 1 and its cases of failure

Condition	a as AE root	PI of e^{ax}	Example
$f(a) \neq 0$	not a root	$\frac{e^{ax}}{f(a)}$	$\frac{e^{x}}{D^2-3D-4} = -\frac{e^{x}}{6}$
$f(a) = 0,\ f'(a) \neq 0$	simple	$x\,\frac{e^{ax}}{f'(a)}$	$\frac{4e^{4x}}{D^2-3D-4} = \frac{4}{5}xe^{4x}$
$f'(a) = 0,\ f''(a) \neq 0$	double	$x^2\,\frac{e^{ax}}{f''(a)}$	$\frac{e^{-2x}}{(D+2)^2} = \frac{x^2}{2}e^{-2x}$
$f''(a) = 0,\ f'''(a) \neq 0$	triple	$x^3\,\frac{e^{ax}}{f'''(a)}$	$\frac{4e^{-x}}{(D+1)^3(D-2)} = -\frac{2}{9}x^3e^{-x}$

S003·8 · S003·13 · T003 00:13:19 · S003·15 · CW U2 P10 · CW U2 P12

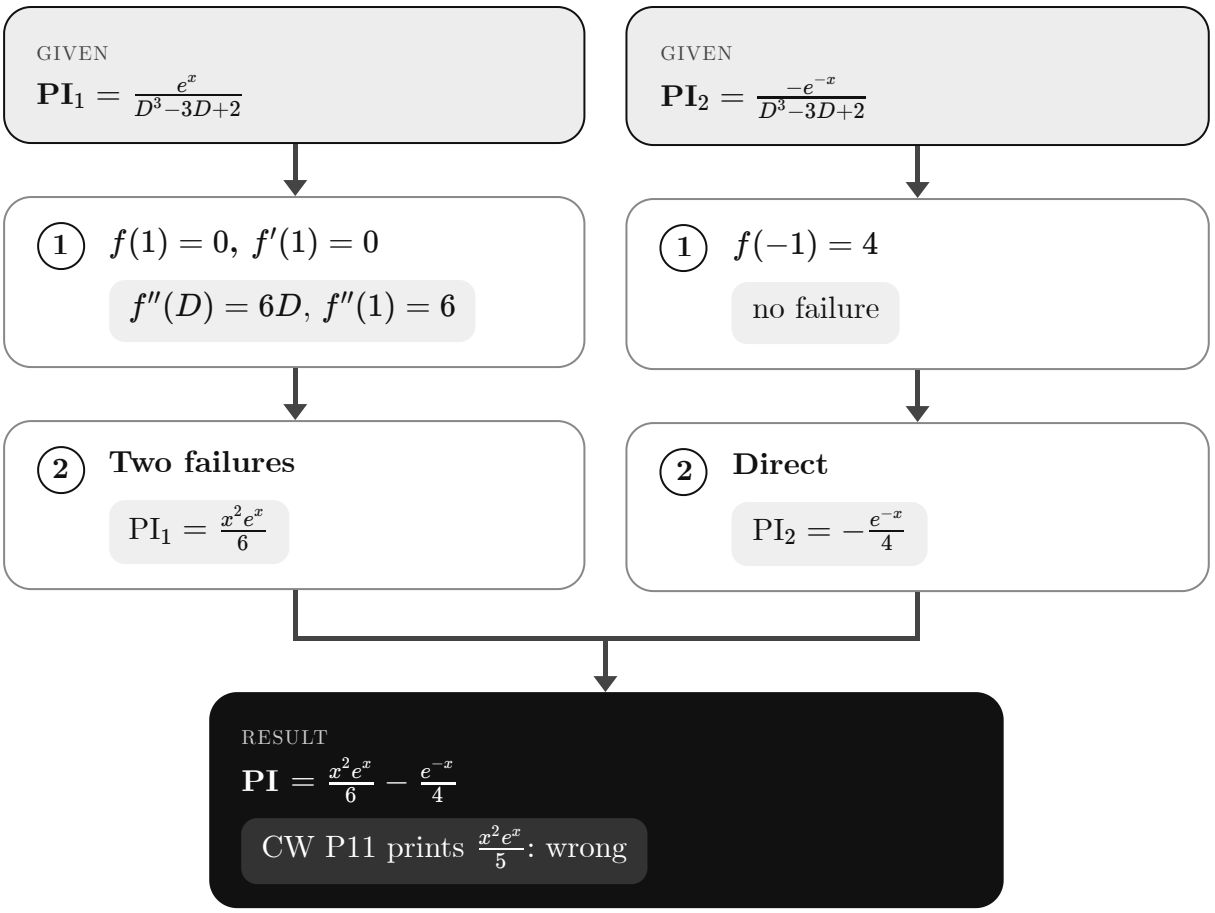
4 Extensions: constants, $\sinh ax$, $\cosh ax$ and a^x



Remark. If the replaced value is a root of f, climb the failure ladder (F3.1) for that term only.

Figure F4.1. What replaces D? ESA ★★★

S003·8 · S003·9 · S003·10 · T003 00:13:50 · T003 00:16:48



Remark. $2\sinh x = e^x - e^{-x}$. The minus sign belongs to PI_2 (slide 003·17 writes $\mathrm{PI}_2 = \frac{e^{-x}}{4}$ and subtracts later).

Figure F4.2. Two terms, two routes: $\frac{2\sinh x}{D^3-3D+2}$ (slides 003·16–17) ESA ★★

S003·16 · S003·17 · T004a 00:05:28 · CW U2 P11

Right-hand sides reducible to Type 1

X	Rewrite as	PI (no failure)
k	ke^{0x}	$\frac{k}{f(0)}$
$\sinh ax$	$\frac{1}{2}(e^{ax} - e^{-ax})$	$\frac{1}{2}\left[\frac{e^{ax}}{f(a)} - \frac{e^{-ax}}{f(-a)}\right]$
$\cosh ax$	$\frac{1}{2}(e^{ax} + e^{-ax})$	$\frac{1}{2}\left[\frac{e^{ax}}{f(a)} + \frac{e^{-ax}}{f(-a)}\right]$
a^x	$e^{(\log a)x}$	$\frac{a^x}{f(\log a)}$
a^{-x}	$e^{-(\log a)x}$	$\frac{a^{-x}}{f(-\log a)}$

S003·9 · S003·10 · T003 00:13:50 · T003 00:16:48

The Type 1 term inside every ESA question

Paper, Q	$f(D)$	Type 1 term	Its PI	Full solution
Dec-23 3a (= May-24 3b, Jun-24 2a)	$(D-2)^2$	$8e^{2x}$	$4x^2e^{2x}$ (W3.5)	class 004
Jul-23 3b	D^2-5D+6	e^{4x}	$\frac{1}{2}e^{4x}$ (W3.6)	class 005
Jan-24 s1 2d	D^2+4	2^{-x}	$\frac{2^{-x}}{(\log 2)^2+4}$ (W4.5)	class 005
Jan-24 s2 2d	$(D-1)(D-2)(D-3)$	e^{2x}	$-xe^{2x}$ (W3.4)	class 005
Jan-24 s3 2c	$D(D-1)^2$	$\frac{1}{2}$ (from $\sin^2 x$)	$\frac{x}{2}$ (W4.7)	class 006
Jan-24 s4 2c	$(D-1)^2(D+1)^2$	$\frac{1}{2}$, 2^{-x}	$\frac{1}{2}+\frac{2^{-x}}{((\log 2)^2-1)^2}$ (W4.6)	class 005

ADDED FOR CLARITY

ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a · ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s3 2c · ESA Jan-24 s4 2c